Assessment Task 3

Data analysis project for marketing campaigns

Subject: 36103 Statistical Thinking for Data Science
Student name: Quang Trung Nguyen
Student ID: 26643559
Date of submission: 1 November 2026
University of Technology Sydney: TD School

Executive summary

The company called 41,168 customers about a new plan, and 11.3% subscribed. I compared four models on 8,234 customers held out from training. Previous subscribers are the best prospects: 65% subscribed again, and they supply 19% of all subscribers. Customers aged 65 and over (47% subscribed) and under 25 (24%) respond far better than those aged 35 to 54 (9%), and mobile calls work better than landline calls.

Calling the top 20% of customers ranked by the logistic regression model would reach 65% of subscribers with 80% fewer calls. I recommend this model, a limit of three call attempts, and a cap on calls to customers aged 65 and over.

Table of contents

Executive summary.....2
Table of contents.....3
Problem formulation.....4
Data and preparation.....5
Methodology.....7
Results.....9
    Model comparison.....9
    Parametric estimates.....10
    Non-parametric findings.....11
    Bayesian estimation.....11
Business insights and strategy.....13
Ethics, privacy and Indigenous data sovereignty.....14
    Consent and privacy.....14
    Fairness, value and risk.....14
    Indigenous data sovereignty.....14
Limitations and conclusion.....15
References.....16
Appendix A. Tests of association with subscription.....17
Appendix B. Logistic regression model selection.....17
Appendix C. Decision tree segments.....18
Appendix D. Bayesian segment estimates.....18
Appendix E. Full odds ratio table (model M4).....19
Appendix F. Code.....21
Appendix G. Bayesian logistic regression compared with maximum likelihood.....22
Appendix H. Declaration of AI use.....23

Problem formulation

The company wants to know which customers respond best to its new plan and how to improve the next campaign, since 36,531 of the 41,168 customers called did not subscribe. This report is for the marketing team that decides whom to call; the models turn customer records into a ranked call list that shows in advance who is likely to say yes. The analysis answers three research questions:

RQ1: Which segments are most likely to subscribe after adjusting for contact method, campaign history and economic conditions?

RQ2: How accurately can subscription be predicted before a call, and does a non-parametric model beat a parametric one?

RQ3: Which calling strategies should the company use in the next campaign?

RQ	Hypothesis	Test
RQ1	H1: Customers aged under 25 and 65+ have higher odds of subscribing than those aged 35-44, after adjustment	Wald z-tests; likelihood-ratio test (LRT) for all age groups together
RQ1	H2: A successful previous campaign is the strongest customer-level predictor	Adjusted odds ratios; permutation importance
RQ1	H3: Cellular contact raises the odds, and the channel effect differs by previous outcome	Wald z-test; LRT for contact type × previous outcome
RQ3	H4: More call attempts in the same campaign lower the odds	Wald z-test for log2(campaign)
RQ2	H5: A non-parametric model ranks customers better than logistic regression	Paired t-test on cross-validation (CV) folds; test AUC; test on campaign phase 3

Table 1. Hypotheses and statistical tests

Data and preparation

Table 2 shows the 11 data issues I found and how I handled each. Chi-square and Mann-Whitney U tests show that every predictor except housing and loan is associated with subscription ($p < 0.05$; Appendix A).

The data has no date column, but the rows are in time order: month runs from May to December, then wraps back to March twice. I call each run of rows with the same month a month block (26 in total), and each wrap starts a new campaign phase, giving three phases of 7, 10 and 9 blocks (Figure 1). Phase 1 averaged 3,954 calls per block at a 4.8% subscription rate; phase 3 averaged 228 calls at 52.1%. Call volume and subscription rate are negatively correlated across blocks (Spearman $\rho = -0.67$, $p < 0.001$).

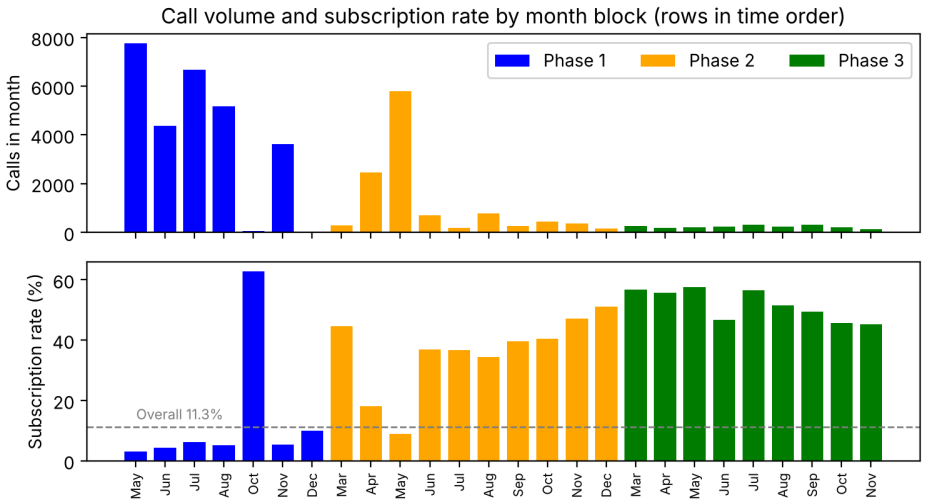


Figure 1. Calls and subscription rate for the 26 month blocks, coloured by phase

Issue	Evidence	Treatment and reason
File format	Each row wrapped in quotes with doubled inner quotes	Repaired before parsing, otherwise the file reads as a single column
Dictionary mismatch	balance and day listed but absent; day_of_week present; pdays uses 999, not −1	Used the actual columns, because only fields that exist can be analysed; differences documented
Duplicates	12 identical rows	Removed, because identical rows add no information and count a customer twice
Target leakage	duration (call length) is known only after the call; cross-validated AUC 0.93 with it vs 0.79 without	Excluded, because the model must predict before the call
pdays	96.3% coded 999 (not contacted before); 4,108 rows have previous > 0 but pdays = 999	Dropped, because poutcome carries the same history without the conflicting code
default	3 'yes' (all non-subscribers), 8,595 'unknown'	Flag default_unk = 1 if default is not ‘no’, because 3 ‘yes’ rows are too few to estimate an effect
housing, loan	Same 990 'unknown' rows; no association (p = 0.06 and 0.58)	Excluded, because neither is associated with subscription
Small category	education = illiterate, n = 18	Merged with basic.4y, because 18 customers are too few for a stable estimate

Issue	Evidence	Treatment and reason
Economic indicators	Variance inflation factor (VIF) up to 64 (euribor3m, emp.var.rate, nr.employed)	Kept nr.employed (number of people employed, quarterly), cons.price.idx (consumer price index) and cons.conf.idx (consumer confidence index), with VIF ≤ 1.4, because collinear predictors make coefficients unstable
Class imbalance	11.3% subscribed	Stratified splits, PR-AUC and a tuned threshold, with no resampling, so predicted probabilities stay calibrated
'unknown' values	Informative: default unknown subscribes at 5.2% vs 12.9% for default = 'no'	Kept as a separate level, because 'unknown' itself carries information

Table 2. Data issues, treatment and reason

Methodology

I split the data 80/20 into training (n = 32,934) and test (n = 8,234) sets, stratified by subscription status, and used only the training set for selection and tuning.

The parametric model is a logistic regression estimated by maximum likelihood (MLE). I built six candidate models (Table 4): M1 to M4 add one block of predictors at a time, and M5 and M6 add an interaction to M4. LRTs, AIC and BIC (Appendix B) selected M4 as the final model. Age enters as six groups (17-24, 25-34, 35-44, 45-54, 55-64 and 65+) because a quadratic term showed a U-shape: young and old customers subscribe more (LRT $p < 0.001$). The largest VIF in M4 is 3.3.

The three non-parametric models (KNN, decision tree and gradient boosting) use the same predictors, so accuracy differences come from the model type. Each was tuned by grid search with 5-fold stratified CV on the training set. Models are scored by ROC-AUC, the chance that a random subscriber is ranked above a random non-subscriber (0.5 is random guessing), because the task is ranking customers.

Besides MLE, I use Bayesian estimation: the posterior is proportional to the likelihood times the prior. Segment subscription rates (for example, by age group) use a Beta prior and Binomial likelihood. I also refitted M4 with Normal(0, 2.5^2) priors and sampled its posterior with MCMC (Gelman et al., 2013), checking convergence with trace, autocorrelation and density plots. A time-based test trains on phases 1 and 2 and predicts phase 3.

Model	Type and reason	Configuration and tuning grid	Selected
Logistic regression	Parametric. Binary outcome; coefficients give odds ratios	MLE (statsmodels; Seabold & Perktold, 2010). Six nested candidate models (Table 4). nr.employed chosen over euribor3m (AIC 18,484 vs 18,504). log2(campaign) gives the OR per doubling of calls.	Model M4 (all four blocks), 47 parameters: intercept + 46 coefficients
KNN	Non-parametric. No functional form; similar customers behave alike	One-hot and standardised inputs. k: 100, 200, 400, 800, 1,600; uniform or distance weights	k = 800, uniform
Decision tree	Non-parametric. Readable segment rules	Depth: 4, 6, 8, 10. Minimum leaf: 50, 100, 200, 400	Depth 6, minimum leaf 100
Gradient boosting	Non-parametric. Learns interactions and curves	Learning rate 0.05, 0.1; depth 2, 3, 5; leaf 100, 300, 1,000; L2 1, 10, 30; early stopping after 30 rounds (max 1,000 trees)	Rate 0.1, depth 3, leaf 300, L2 30, 535 trees. All 54 settings within 0.005 CV AUC
Beta-Binomial (Bayesian)	Bayesian estimation. Uncertainty for small segments	Prior Beta(2.25, 17.75): mean 11.3%, weight of 20 customers. Sensitivity prior Beta(1, 1)	Posterior mean and 95% credible interval
Bayesian logistic regression	Bayesian estimation. Posterior for the 47 coefficients of M4; no closed form, so MCMC	Normal(0, 2.5^2) priors on standardised predictors. Random-walk Metropolis-Hastings, one chain of 60,000 iterations, burn-in 5,000, keep every 50th draw. Sensitivity prior Normal(0, 0.5^2)	1,100 draws; acceptance rate 0.24; converged (Figure 7)

Table 3. Model choice, configuration and training

Note. All classifiers: 5-fold stratified CV on the training set, ROC-AUC scoring, F1-optimal threshold from out-of-fold predictions, random seed 42. Models were fitted with scikit-learn (Pedregosa et al., 2011); model definitions follow James et al. (2021).

Model	Predictors	Parameters	AIC	Result
M1	Customer profile: age group, job, marital, education, default_unk	27	21,961.2	Baseline
M2	M1 + contact history: contact, poutcome, log2(campaign)	31	20,142.5	Better than M1 (LRT $p < 0.001$)
M3	M2 + economy: nr.employed, cons.price.idx, cons.conf.idx	34	18,816.4	Better than M2 (LRT $p < 0.001$)
M4	M3 + timing: month, day_of_week	47	18,484.4	Better than M3 (LRT $p < 0.001$); final model
M5	M4 + interaction contact × poutcome	49	18,488.3	No improvement (LRT $p = 0.94$)
M6	M4 + interaction contact × age group	52	18,487.1	No improvement (LRT $p = 0.20$)

Table 4. Candidate logistic regression models, fitted on the training set (lower AIC is better)

Results

Model comparison

All four models rank customers well (Table 5, Figure 2): test ROC-AUC is 0.797 to 0.807, and PR-AUC is about four times the no-skill value of 0.113. Gradient boosting beats logistic regression in CV by 0.008 AUC (paired t-test p = 0.013), but on the test set the gap is only 0.007 (0.807 vs 0.800). The top 20% ranked by either of these two models hold 65% to 66% of subscribers (Figure 3).

In the time-based test, ROC-AUC falls from about 0.80 to between 0.60 and 0.70, because nr.employed in phase 3 is lower than any value seen in training. Models using only customer and contact variables reach 0.70, so customer patterns carry over to a later phase and economic patterns do not.

Model	CV ROC-AUC (sd)	CV PR-AUC (sd)	Test ROC-AUC	Test PR-AUC	Precision	Recall	F1	Top-20% capture	Phase-3 ROC-AUC
Logistic regression	0.784 (0.011)	0.439 (0.017)	0.800	0.462	0.448	0.586	0.508	0.648	0.674
KNN	0.783 (0.008)	0.407 (0.018)	0.799	0.434	0.448	0.578	0.505	0.652	0.602
Decision tree	0.780 (0.010)	0.427 (0.021)	0.797	0.450	0.464	0.586	0.518	0.635	0.697
Gradient boosting	0.792 (0.008)	0.455 (0.022)	0.807	0.477	0.472	0.591	0.525	0.660	0.647

Table 5. Model performance (CV on training set; test set n = 8,234; phase-3 test n = 2,054)

Note. Phase-3 models were trained on phases 1 and 2. Customer and contact variables only: logistic regression 0.697, gradient boosting 0.698. PR-AUC summarises precision across all recall levels; its no-skill value equals the 11.3% subscription rate. Precision, recall and F1 use the F1-optimal threshold from CV. Top-20% capture is the share of all subscribers found among the 20% of test customers with the highest scores.

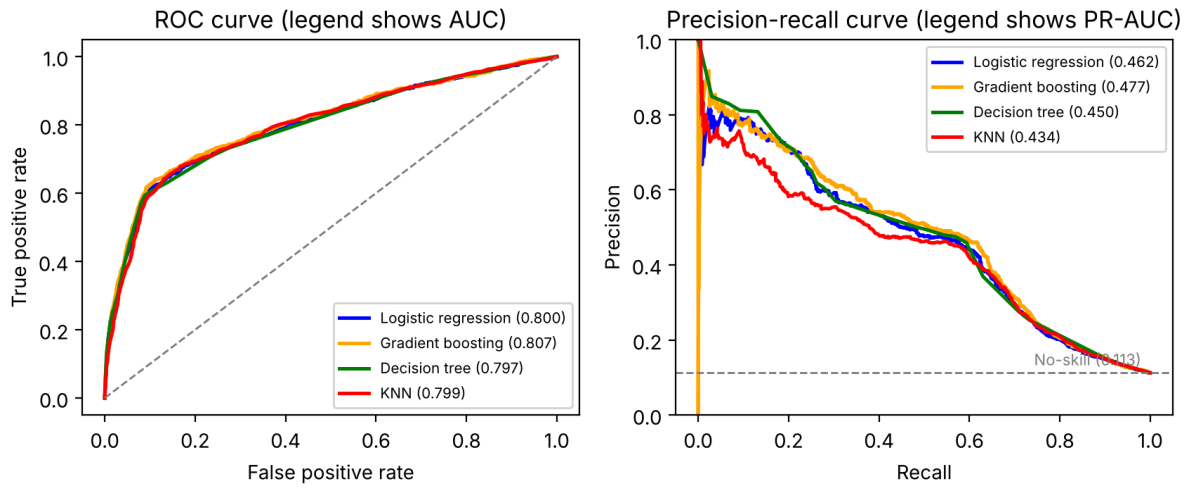


Figure 2. ROC and precision-recall curves on the test set

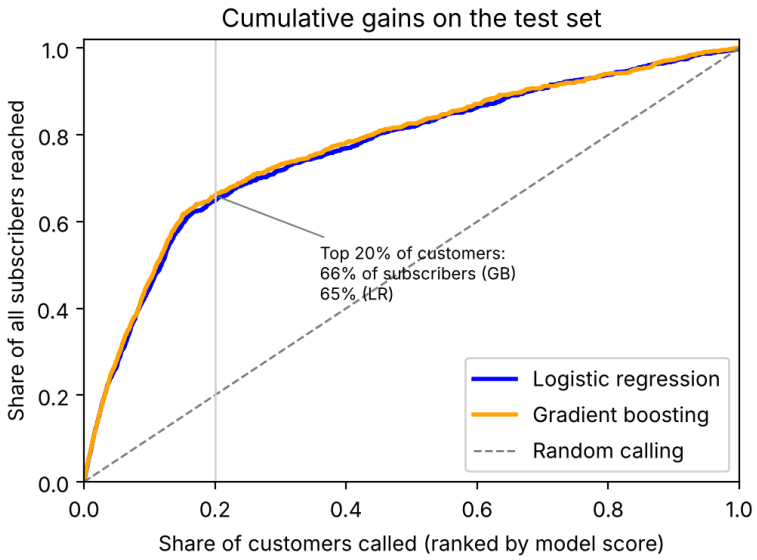


Figure 3. Cumulative gains: share of subscribers reached by calling the top-ranked customers

Parametric estimates

Figure 4 shows adjusted odds ratios (OR) from M4; an OR above 1 means higher odds than the reference group, with other variables held fixed. A previous success raises the odds 3.68 times (95% CI 3.17 to 4.29) compared with never-contacted customers. A previous failure lowers them (OR 0.63) although its raw rate is higher (14.2% vs 8.8%), because 84% of previous failures were called in phases 2 and 3, when overall rates were high. Telephone (landline) contact cuts the odds by 38% (OR 0.62), and each doubling of call attempts cuts them by 7% (OR 0.93).

Compared with ages 35-44, the odds are 1.48 times higher at 65+ and 1.30 times higher at 17-24. Once age is in the model, the student and retired effects are not significant (p = 0.25 and 0.18), so their high raw rates (31% and 25%) mainly reflect age. M5 and M6 did not improve on M4 (LRT p = 0.94 and 0.20).

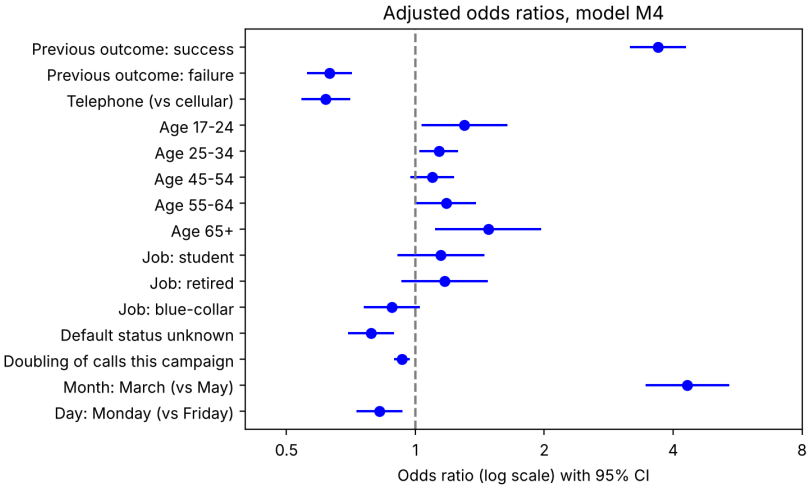


Figure 4. Selected adjusted odds ratios from logistic model M4 (reference levels: age 35-44, admin., cellular, never contacted, May, Friday)

Non-parametric findings

Permutation importance ranks nr.employed far above every customer variable (Figure 5). Partial dependence on age shows the same U-shape as the logistic model. The best tree segment is previous subscribers called when nr.employed was below 5,088, with subscription rates of 64% to 77% (Appendix C).

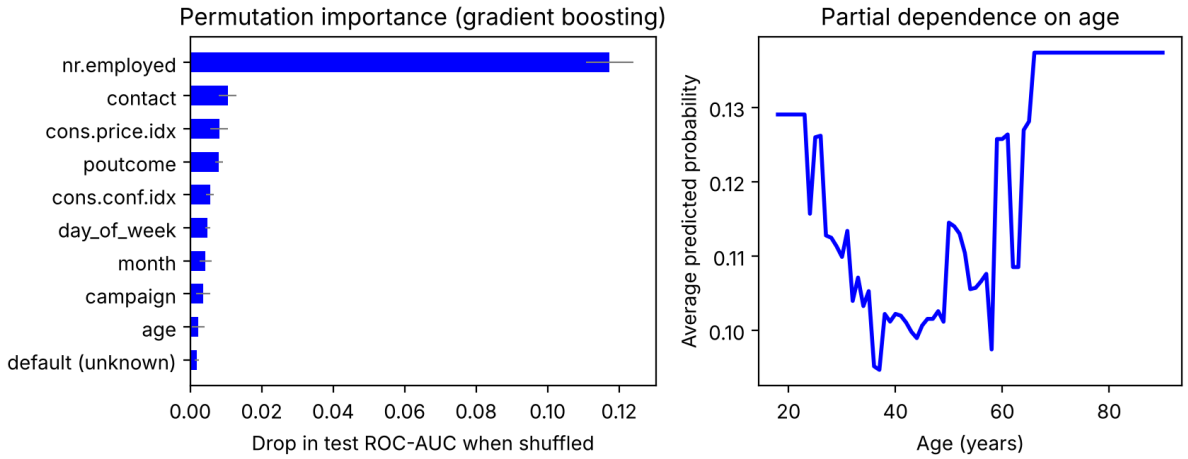


Figure 5. Gradient boosting: permutation importance on the test set (drop in AUC when one variable is shuffled; left) and partial dependence on age (average predicted probability at each age; right)

Bayesian estimation

Customers aged 65+ have a 95% credible interval of 42.6% to 50.0% for their rate (Figure 6). The prior matters most for small segments: for the 18 illiterate customers it pulls the observed 22.2% toward the overall 11.3%, giving 16.5% (6.6% to 29.6%).

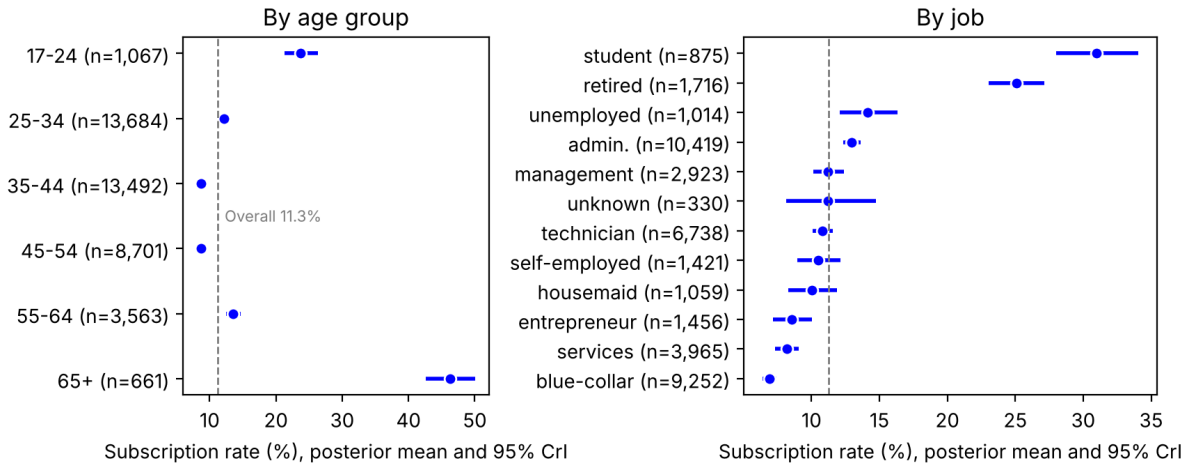


Figure 6. Posterior subscription rate by age group and job (Beta-Binomial, 95% credible intervals)

The Bayesian logistic regression converged: the trace plots are stable after the burn-in, and thinning removes most of the autocorrelation (Figure 7). Its odds ratios match the MLE within 0.11 standard errors, and a tighter prior moves no coefficient by more than 1.1 standard errors (Appendix G), so the data outweigh the prior. Unlike a confidence interval, a credible interval gives probabilities directly: the odds ratio for a previous success exceeds 3 with probability 0.997.

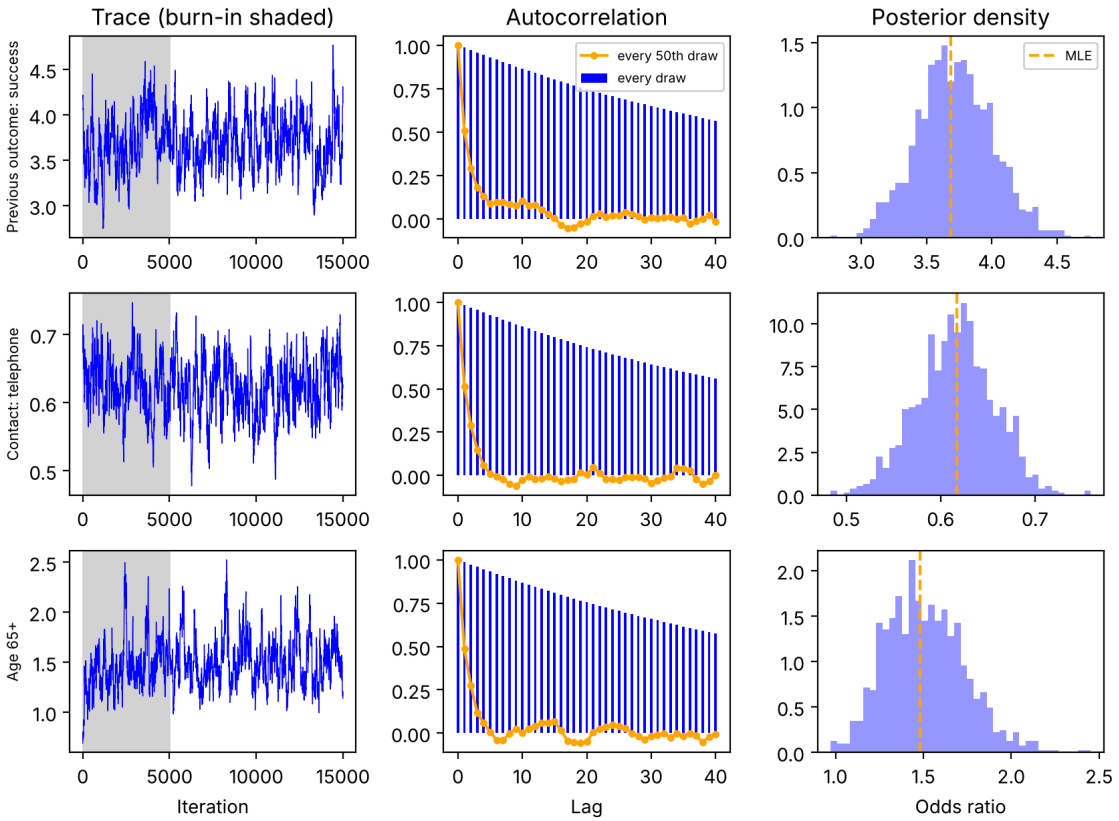


Figure 7. MCMC diagnostics for three odds ratios: trace of the chain (burn-in shaded), autocorrelation of every draw and of every 50th draw, and posterior density (dashed line: maximum likelihood estimate)

Hypothesis	Evidence	Decision
H1 age	17-24: OR 1.30 [1.03, 1.64]; 65+: OR 1.48 [1.11, 1.97]; age block LRT p = 0.018	Supported
H2 previous success	OR 3.68 [3.17, 4.29], the largest customer-level effect; `nr.employed` dominates overall importance	Supported at customer level
H3 channel	Telephone OR 0.62 [0.54, 0.70]; interaction LRT p = 0.94	Main effect supported; interaction not
H4 call attempts	OR 0.93 [0.89, 0.97] per doubling	Supported
H5 model type	CV gain +0.008 (p = 0.013); test gain only +0.007; phase-3 AUC 0.647 vs 0.674	Not supported in practice

Table 6. Hypothesis outcomes

Business insights and strategy

Call previous subscribers first: they are 3.3% of customers but 19.2% of subscribers (Table 7). Design offers for each age group, since job adds little once age is known. Call mobiles first and stop after three attempts: customers needing more than three calls are 18.5% of customers but 12.0% of subscribers.

I recommend the logistic model for deployment, refitted before each campaign. Its test AUC is within 0.007 of gradient boosting, it held up better in phase 3 (0.674 vs 0.647), and its coefficients can be audited.

Insight	Evidence	Action	Measure
1. Previous subscribers convert best	65.1% rate; 3.3% of customers, 19.2% of subscribers; OR 3.68	Call this list first in every campaign	Conversion of the re-contact list
2. Age drives response more than job	65+: 47%, OR 1.48; 17-24: 24%, OR 1.30; job effects not significant after age	Design offers and scripts by age group	Conversion and complaints by age group
3. Mobile beats landline; extra attempts lose value	Telephone OR 0.62; 13.0% at first call vs 5.5% at six or more; OR 0.93 per doubling	Call mobiles first; stop after three attempts	Calls per subscription
4. Mass calling converts poorly	Phase 1: 3,954 calls per month block at 4.8%; $\rho = -0.67$ between volume and rate	Call the model's top 20% to 30% instead of everyone	Share of subscribers reached
5. Economic effects do not carry forward	Phase-3 AUC 0.60-0.70 vs 0.80 on random test; `nr.employed` outside training range	Refit before each campaign; check AUC monthly	AUC on the latest month

Table 7. Insights, evidence and recommended actions

Ethics, privacy and Indigenous data sovereignty

Consent and privacy

The data links personal and credit details to call outcomes, so Australian Privacy Principle 7 (APP 7, direct marketing) in the Privacy Act 1988 applies: direct marketing needs customer consent or a reasonable expectation, and each campaign needs a simple opt-out (OAIC, 2019). The dataset has no consent or opt-out record, so the company cannot show that customers agreed. From 10 December 2026, privacy policies must also disclose computer programs that substantially and directly support decisions that significantly affect individuals (Privacy and Other Legislation Amendment Act 2024). A model that decides who receives offers may meet this test.

Fairness, value and risk

Logistic regression and gradient boosting both put 98.6% of test customers aged 65+ on the top-20% call list, so a group that makes up 1.8% of customers would receive 8.7% of calls. A high response rate does not show that the plan suits them, and older customers can be more exposed to pressure on the phone. The model also reduces harm: calling the top 30% avoids 70% of calls, mostly to people who would say no. Default status adds little accuracy (permutation importance 0.002), so I recommend removing it.

Indigenous data sovereignty

The data has no field identifying Aboriginal and Torres Strait Islander customers, so nobody can check how the model treats them. About 43% of the 1,545 First Nations communities and homelands in Australia have no mobile service (RMIT University, 2023). A model that favours mobile (cellular) contact would rank them low without anyone noticing.

Adding an Indigenous identifier to the data would not fix this on its own. The Maiam nayri Wingara principles give Aboriginal and Torres Strait Islander people control of the data ecosystem and the right to data that support self-determination (Maiam nayri Wingara & AIGI, 2018). The CARE principles add collective benefit and authority to control (Carroll et al., 2020). Such data should be collected only with free, prior and informed consent, governed with Aboriginal community-controlled organisations, and reported back to communities (Table 8). I would also avoid building proxies for Indigeneity from location or contact type, because group-level inferences can harm a community even when individuals are anonymous.

Risk	Evidence	Control	Owner
No consent or opt-out record	Dataset has no consent field	Record consent and opt-outs; remove opted-out customers before scoring	Privacy officer
Automated decision not disclosed	Model ranks who receives offers	Describe the model in the privacy policy before 10 December 2026; keep model documentation	Legal and analytics
Over-targeting older customers	98.6% of customers aged 65+ selected	Cap the share of calls per age group; cooling-off period; track complaints and cancellations by age	Marketing manager

Risk	Evidence	Control	Owner
Unneeded financial data	`default_unk` importance 0.002	Remove from the deployed model	Analytics team
First Nations customers invisible	No identifier; 43% of 1,545 First Nations communities have no mobile service	Do not infer Indigeneity; co-design any data collection with community-controlled organisations under Maiam nayri Wingara and CARE; report results back	Executive sponsor

Table 8. Ethical risks and recommended controls for business stakeholders and end users

Limitations and conclusion

The variables match public bank telemarketing data collected in Portugal from 2008 to 2010 (Moro et al., 2014), so results may not transfer to current Australian customers. The models predict who says yes on a call; they cannot show that the call caused the subscription.

The most responsive segments are previous subscribers, customers under 25 or over 65, and customers reached by mobile (RQ1; Table 6). The logistic model ranks customers before a call with test AUC 0.80, but AUC falls to 0.67 on a later phase (RQ2). For RQ3, the company should call previous subscribers first, prefer mobiles, stop after three attempts and call only the model's top 20% to 30%. The next step is a one-month trial comparing a model-ranked list with an equal random list.

References

Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE principles for Indigenous data governance. *Data Science Journal, 19*(1), 43. https://doi.org/10.5334/dsj-2020-043

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman and Hall/CRC.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. https://doi.org/10.1007/978-1-0716-1418-1

Maiam nayri Wingara Indigenous Data Sovereignty Collective, & Australian Indigenous Governance Institute. (2018). *Indigenous data sovereignty communique: Indigenous Data Sovereignty Summit, 20 June 2018, Canberra*. https://www.aigi.org.au/wp-content/uploads/2022/01/Communique-Indigenous-Data-Sovereignty-Summit-1.pdf

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems, 62*, 22-31. https://doi.org/10.1016/j.dss.2014.03.001

Office of the Australian Information Commissioner. (2019). *Chapter 7: APP 7 Direct marketing* (Australian Privacy Principles guidelines, Version 1.1). https://www.oaic.gov.au/privacy/australian-privacy-principles/australian-privacy-principles-guidelines/chapter-7-app-7-direct-marketing

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825-2830.

Privacy Act 1988 (Cth).

Privacy and Other Legislation Amendment Act 2024 (Cth).

RMIT University. (2023, September 27). *Remote First Nations communities are among the most digitally excluded people in Australia: Report* [Media release]. https://www.rmit.edu.au/news/all-news/2023/sep/mapping-digital-gap

Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference* (pp. 92-96). https://doi.org/10.25080/Majora-92bf1922-011

Appendix A. Tests of association with subscription

Variable	Test	Statistic	df	p-value	Effect size
poutcome	Chi-square	4217.7	2	<0.001	0.32
month	Chi-square	3103.7	9	<0.001	0.275
job	Chi-square	960.0	11	<0.001	0.153
contact	Chi-square	860.8	1	<0.001	0.145
default	Chi-square	406.2	2	<0.001	0.099
education	Chi-square	192.4	7	<0.001	0.068
marital	Chi-square	123.2	3	<0.001	0.055
day_of_week	Chi-square	26.0	4	<0.001	0.025
housing	Chi-square	5.8	2	0.056	0.012
loan	Chi-square	1.1	2	0.584	0.005
age	Mann-Whitney U	82,840,420	-	0.015	0.022
campaign	Mann-Whitney U	75,347,760	-	<0.001	0.11
euribor3m	Mann-Whitney U	43,474,104	-	<0.001	0.487
nr.employed	Mann-Whitney U	42,537,496	-	<0.001	0.498
cons.price.idx	Mann-Whitney U	65,941,150	-	<0.001	0.221
cons.conf.idx	Mann-Whitney U	91,039,638	-	<0.001	0.075

Table A1. Chi-square tests (Cramér's V) and Mann-Whitney U tests (rank-biserial r), full cleaned data (n = 41,168)

Appendix B. Logistic regression model selection

Model	Parameters	Log-likelihood	AIC	BIC	McFadden R^2
M1 customer profile	27	-10,953.6	21,961.2	22,188.1	0.055
M2 + contact history	31	-10,040.2	20,142.5	20,402.9	0.134
M3 + economy	34	-9,374.2	18,816.4	19,102.1	0.191
M4 + timing	47	-9,195.2	18,484.4	18,879.3	0.207
M4 using euribor3m	47	-9,205.2	18,504.4	18,899.3	0.206
M5 + contact x previous outcome	49	-9,195.1	18,488.3	18,900.0	0.207
M6 + contact x age group	52	-9,191.6	18,487.1	18,924.1	0.207
M4 without age	42	-9,202.0	18,488.0	18,840.9	0.206
M4 with linear age	43	-9,202.0	18,489.9	18,851.2	0.206
M4 with age + age^2	44	-9,195.3	18,478.5	18,848.2	0.207

Table B1. Nested logistic models fitted by maximum likelihood on the training set (n = 32,934)

Note. M1 to M6 are defined in Table 4. "M4 using euribor3m" replaces nr.employed with euribor3m. The last three rows change only how age enters M4: left out, as a straight line, or as a curve (age + age²). Lower AIC and BIC mean a better balance between fit and number of parameters; McFadden R^2 is a pseudo R^2 for logistic models.

Comparison	LR χ^2	df	p-value
M1 customer profile → M2 + contact history	1826.8	4	< 0.001
M2 + contact history → M3 + economy	1332.1	3	< 0.001
M3 + economy → M4 + timing	358.0	13	< 0.001
M4 + timing → M5 + contact x previous outcome	0.1	2	0.940
M4 + timing → M6 + contact x age group	7.3	5	0.201
M4 without age → M4 + timing	13.6	5	0.018
M4 with linear age → M4 with age + age^2	13.4	1	< 0.001

Table B2. Likelihood-ratio tests

Appendix C. Decision tree segments

Segment rule	Customers	Share of customers	Subscription rate	Share of subscribers
nr.employed <= 5,087.7 and poutcome = success and cons.price.idx > 93.2	523	1.6%	76.7%	10.8%
nr.employed <= 5,087.7 and poutcome = success and cons.price.idx <= 93.2	380	1.2%	64.2%	6.6%
nr.employed <= 5,087.7 and poutcome != success and contact = cellular	2,602	7.9%	38.5%	27.0%
nr.employed > 5,087.7 and cons.conf.idx <= -46.6 and poutcome != failure	1,673	5.1%	23.6%	10.6%
nr.employed <= 5,087.7 and poutcome != success and contact != cellular	482	1.5%	22.6%	2.9%
nr.employed > 5,087.7 and cons.conf.idx <= -46.6 and poutcome = failure	496	1.5%	11.1%	1.5%
nr.employed > 5,087.7 and cons.conf.idx > -46.6 and cons.price.idx <= 94.0	17,096	51.9%	6.8%	31.3%
nr.employed > 5,087.7 and cons.conf.idx > -46.6 and cons.price.idx > 94.0	9,682	29.4%	3.6%	9.3%

Table C1. Leaves of a depth-3 decision tree (minimum 300 customers per leaf), training set

Appendix D. Bayesian segment estimates

Segment	n	Observed	Posterior mean	95% credible interval	P(rate > 11.3%)	Uniform prior 95% CrI
Age: 17-24	1,067	24.0%	23.8%	21.3% to 26.3%	1.000	0.215 to 0.266
Age: 25-34	13,684	12.2%	12.2%	11.6% to 12.7%	1.000	0.116 to 0.127
Age: 35-44	13,492	8.6%	8.7%	8.2% to 9.1%	0.000	0.082 to 0.091
Age: 45-54	8,701	8.6%	8.6%	8.1% to 9.2%	0.000	0.081 to 0.093
Age: 55-64	3,563	13.6%	13.5%	12.4% to 14.7%	1.000	0.125 to 0.147
Age: 65+	661	47.4%	46.3%	42.6% to 50.0%	1.000	0.436 to 0.512
Job: blue-collar	9,252	6.9%	6.9%	6.4% to 7.4%	0.000	0.064 to 0.074
Job: retired	1,716	25.2%	25.1%	23.1% to 27.1%	1.000	0.232 to 0.273

Segment	n	Observed	Posterior mean	95% credible interval	P(rate > 11.3%)	Uniform prior 95% CrI
Job: student	875	31.4%	31.0%	28.0% to 34.0%	1.000	0.284 to 0.346
Education: illiterate	18	22.2%	16.5%	6.6% to 29.6%	0.802	0.091 to 0.456
Education: university.degree	12,162	13.7%	13.7%	13.1% to 14.3%	1.000	0.131 to 0.143
Previous outcome: failure	4,250	14.2%	14.2%	13.2% to 15.3%	1.000	0.132 to 0.153
Previous outcome: nonexistent	35,547	8.8%	8.8%	8.5% to 9.1%	0.000	0.085 to 0.091
Previous outcome: success	1,371	65.1%	64.3%	61.8% to 66.8%	1.000	0.625 to 0.675
Contact: cellular	26,131	14.7%	14.7%	14.3% to 15.2%	1.000	0.143 to 0.152
Contact: telephone	15,037	5.2%	5.2%	4.9% to 5.6%	0.000	0.049 to 0.056

Table D1. Posterior subscription rates (prior Beta(2.25, 17.75)); last column uses the uniform prior Beta(1, 1)

Appendix E. Full odds ratio table (model M4)

Term	OR	95% CI	p-value
Age 17-24	1.302	1.034 to 1.639	0.025
Age 25-34	1.134	1.023 to 1.258	0.017
Age 45-54	1.094	0.972 to 1.231	0.135
Age 55-64	1.179	1.004 to 1.386	0.045
Age 65+	1.479	1.112 to 1.968	0.007
Job: blue-collar	0.881	0.758 to 1.024	0.100
Job: entrepreneur	1.007	0.799 to 1.269	0.955
Job: housemaid	0.927	0.701 to 1.225	0.592
Job: management	0.978	0.830 to 1.152	0.788
Job: retired	1.170	0.928 to 1.476	0.185
Job: self-employed	0.934	0.747 to 1.169	0.553
Job: services	0.886	0.753 to 1.043	0.146
Job: student	1.146	0.907 to 1.448	0.254
Job: technician	0.996	0.870 to 1.140	0.950
Job: unemployed	0.871	0.676 to 1.121	0.284
Job: unknown	1.128	0.729 to 1.746	0.590
Marital: married	1.084	0.950 to 1.237	0.233
Marital: single	1.107	0.953 to 1.286	0.185
Marital: unknown	1.373	0.635 to 2.967	0.420
Education: basic.4y	0.898	0.753 to 1.071	0.230
Education: basic.6y	1.056	0.856 to 1.302	0.610
Education: basic.9y	0.852	0.729 to 0.995	0.044
Education: high.school	0.936	0.834 to 1.050	0.257
Education: professional.course	0.939	0.813 to 1.084	0.387
Education: unknown	1.058	0.867 to 1.291	0.581
Contact: telephone	0.617	0.541 to 0.704	< 0.001
Previous outcome: failure	0.631	0.559 to 0.711	< 0.001
Previous outcome: success	3.684	3.166 to 4.287	< 0.001
Month: apr	1.997	1.730 to 2.305	< 0.001

Term	OR	95% CI	p-value
Month: aug	1.650	1.397 to 1.948	< 0.001
Month: dec	2.296	1.586 to 3.324	< 0.001
Month: jul	2.319	1.990 to 2.701	< 0.001
Month: jun	2.478	2.141 to 2.868	< 0.001
Month: mar	4.315	3.444 to 5.407	< 0.001
Month: nov	1.389	1.182 to 1.632	< 0.001
Month: oct	1.592	1.274 to 1.990	< 0.001
Month: sep	1.154	0.901 to 1.477	0.256
Day: mon	0.824	0.727 to 0.933	0.002
Day: thu	1.017	0.901 to 1.148	0.787
Day: tue	0.999	0.881 to 1.133	0.989
Day: wed	1.123	0.993 to 1.271	0.064
default_unk (default not 'no')	0.788	0.695 to 0.893	< 0.001
log2(campaign)	0.930	0.891 to 0.971	< 0.001
nr.employed (per 1 unit)	0.990	0.989 to 0.990	< 0.001
cons.price.idx	0.982	0.904 to 1.066	0.667
cons.conf.idx	1.020	1.009 to 1.030	< 0.001

Table E1. Odds ratios with 95% confidence intervals (reference levels: age 35-44, admin., divorced, university.degree, cellular, nonexistent, May, Friday)

Appendix F. Code

All analysis code is in the notebook QuangTrungNguyen_Assessment3_Code_01112026.ipynb, submitted with this report. It runs from top to bottom with TeleCom_Data-2.csv in the same folder (random seed 42) and shows every table and figure inside the notebook without writing any files. Sections: 1 setup and loading; 2 data checks and cleaning; 3 tests of association; 4 train/test split; 5 logistic regression (MLE, model selection, assumptions, odds ratios); 6 non-parametric models and tuning; 7 model comparison (CV, test set, phase-3 test); 8 permutation importance, partial dependence and tree segments; 9 Bayesian estimation (segment rates; logistic regression by MCMC with diagnostics and prior sensitivity); 10 business insight calculations; 11 fairness check; 12 hypothesis summary.

Appendix G. Bayesian logistic regression compared with maximum likelihood

Term	MLE OR	95% CI	Bayes OR	95% CrI
Previous outcome: success	3.684	3.166 to 4.287	3.712	3.161 to 4.299
Previous outcome: failure	0.631	0.559 to 0.711	0.630	0.555 to 0.710
Contact: telephone	0.617	0.541 to 0.704	0.617	0.534 to 0.691
Age 17-24	1.302	1.034 to 1.639	1.297	1.025 to 1.625
Age 25-34	1.134	1.023 to 1.258	1.132	1.022 to 1.262
Age 45-54	1.094	0.972 to 1.231	1.091	0.967 to 1.234
Age 55-64	1.179	1.004 to 1.386	1.180	0.998 to 1.377
Age 65+	1.479	1.112 to 1.968	1.493	1.128 to 2.002
Job: student	1.146	0.907 to 1.448	1.148	0.919 to 1.477
Job: retired	1.170	0.928 to 1.476	1.174	0.924 to 1.455
Job: blue-collar	0.881	0.758 to 1.024	0.887	0.765 to 1.020
default_unk (default not 'no')	0.788	0.695 to 0.893	0.784	0.691 to 0.893
log2(campaign)	0.930	0.891 to 0.971	0.930	0.892 to 0.970
nr.employed (per 1 unit)	0.990	0.989 to 0.990	0.990	0.989 to 0.990
Month: mar	4.315	3.444 to 5.407	4.350	3.455 to 5.417
Day: mon	0.824	0.727 to 0.933	0.826	0.731 to 0.940

Table G1. Odds ratios from maximum likelihood (95% CI) and from the Bayesian model (posterior median and 95% credible interval)

Note. Normal(0, 2.5²) priors; one chain, 1,100 retained draws. All 46 terms are listed in Section 9.2.2 of the notebook. With the tighter Normal(0, 0.5²) prior the largest shift in a posterior mean is 1.1 standard errors (median 0.24); for the terms in Table 7 it is 0.6 or less.

Appendix H. Declaration of AI use

I used two AI tools while preparing this assessment. The final analysis decisions, model choices, interpretations and recommendations are my own, and I checked every AI output before using it.

Claude (Anthropic) was used at the start to suggest possible directions for the analysis. After I chose a direction, I asked it to grade my drafts against the rubric and give feedback, and I refined my ideas over several rounds of this. It was also used to check spelling and grammar, to shorten sentences so the report fits the word limit, and to restructure and tidy the notebook (code layout, comments and chart formatting) so it is easier to read. I decided which suggestions to accept, and I reviewed and edited all text and code before including it.

NotebookLM (Google) was used to search for relevant literature and to check that the facts and citations I used match the original sources.

All numbers, tables and figures in this report come from the code in the submitted notebook.